ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Chemical Reactions and Equations** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 12	$3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$: 3+4+1+4=12.
2	**(d)** Fe_2O_3	Fe_2O_3 loses oxygen.
3	**(a)** $FeSO_4$ (pale green) is formed	Iron displaces copper.
4	**(a)** 5	$C_3H_8 + 5\,O_2 \longrightarrow 3\,CO_2 + 4\,H_2O$.
5	**(a)** C	Carbon takes oxygen from ZnO.
6	**(b)** water contains 2 hydrogen atoms for each oxygen atom	$2\,H_2O \longrightarrow 2\,H_2 + O_2$.
7	**(c)** 3	$2\,KClO_3 \longrightarrow 2\,KCl + 3\,O_2$.
8	**(d)** thermal decomposition	Heat causes the break-down.
9	**(c)** prevent oxidation of the oil in the chips	Nitrogen is inert.
10	**(c)** 9	$4\,Al + 3\,O_2 \longrightarrow 2\,Al_2O_3$: 4+3+2=9.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Physical change: no new substance, e.g. melting of ice (1); Chemical change: new substance formed, e.g. rusting of iron (1).	1+1
12	$2\,Al + 3\,H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 3\,H_2$ (1); Hydrogen gas (1).	1+1
13	They decompose in the presence of light (1); $2\,AgBr \longrightarrow 2\,Ag + Br_2$ (1).	1+1
14	It releases heat energy (1); $CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O + \text{heat}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Al on LHS 1, RHS 2: put 2 before Al (1); $2\,Al + 3\,H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 3\,H_2$ (1); Displacement reaction (1).	1+1+1
16	Reaction is endothermic (1); $CaCO_3 \longrightarrow CaO + CO_2$ (1); Quick lime is used in cement/whitewash (1).	1+1+1
17	Cathode: hydrogen; anode: oxygen (1); Volume ratio 2 : 1 (1); $2\,H_2O \longrightarrow 2\,H_2 + O_2$ (1).	1+1+1
	OR: Balance N and H (1); $4\,NH_3 + 5\,O_2 \longrightarrow 4\,NO + 6\,H_2O$ (1); Oxidation (redox) of ammonia / combustion (1).	1+1+1
18	Blue colour fades; reddish-brown copper deposits (1); $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ (1); Zinc is more reactive than copper (1).	1+1+1
	OR: Crackling sound (1); Brown fumes of NO_2 (1); Yellow residue of PbO (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) Displacement (1); (b) Decomposition (1); (c) Double displacement (1); (d) Combination (1); (e) Displacement (1).	1+1+1+1+1
	OR: Thermal: $CaCO_3 \longrightarrow CaO + CO_2$ (1); Electrolytic: $2\,H_2O \longrightarrow 2\,H_2 + O_2$ (1); Photolytic: $2\,AgBr \longrightarrow 2\,Ag + Br_2$ (1); Photolytic used in photography (1); All need energy, so they are endothermic (1).	1+1+1+1+1
20	(i) Oxidised: H_2; reduced: CuO (1); (i) Oxidising agent CuO; reducing agent H_2 (1); (ii) Oxidised: CO (1); (ii) Reduced: Fe_2O_3 (1); (ii) Oxidising agent Fe_2O_3; reducing agent CO (1).	1+1+1+1+1
	OR: Corrosion: iron combines with oxygen and water (1); Rancidity: fats/oils combine with oxygen (1); Respiration: glucose is oxidised (1); Burning: fuel combines with oxygen (1); All are oxidation reactions that release energy (respiration, burning) or damage (corrosion, rancidity) (1).	1+1+1+1+1

Mistakes to avoid:

- Changing the subscripts (formulae) instead of the coefficients while balancing; H_2O must not be written as H_2O_2 to balance oxygen.
- Writing $Mg + O_2 \longrightarrow MgO$ without balancing, or MgO_2 for magnesium oxide.
- Calling every reaction in which heat is given out 'decomposition'; identify the type from the reactants and products (one product = combination).
- Forgetting that in a displacement reaction the displacing metal must be **more reactive**; $Cu + FeSO_4$ gives no reaction.

- Mixing up oxidation and reduction: gain of oxygen is oxidation, loss of oxygen is reduction.
- Writing that rusting needs only oxygen; it needs **both** oxygen and moisture. Slaking of lime is exothermic, not endothermic.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Chemical Reactions and Equations** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** $CaO + H_2O \longrightarrow Ca(OH)_2$	Quick lime slaking releases heat.
2	**(c)** is more reactive than iron and forms a protective layer	Zinc protects the iron.
3	**(a)** 3	$Fe_2O_3 + 3\,CO \longrightarrow 2\,Fe + 3\,CO_2$.
4	**(a)** HCl	HCl loses hydrogen to give Cl_2.
5	**(a)** Ag_2S formation by H_2S	Sulphur compounds react with silver.
6	**(c)** 7	$2\,C_2H_6 + 7\,O_2 \longrightarrow 4\,CO_2 + 6\,H_2O$.
7	**(a)** CuO	CuO supplies oxygen to H_2.
8	**(b)** Thermal decomposition of $CaCO_3$	Heat must be supplied to decompose limestone.
9	**(b)** 6	$4\,NH_3 + 5\,O_2 \longrightarrow 4\,NO + 6\,H_2O$.
10	**(a)** Fe	Fe is more reactive than Cu; others are less reactive.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Iron displaces copper from copper sulphate (1); Pale green iron sulphate forms (1).	1+1
12	$Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2\,NaCl$ (1); Ions are exchanged between the reactants (1).	1+1
13	Bonds in the compound must be broken, which needs energy (1); Heat, light or electricity (1).	1+1
14	Oxidising agent gives oxygen/removes hydrogen and gets reduced (1); Reducing agent gains oxygen/gives hydrogen and gets oxidised (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Nitrate and hydrogen counts (1); $Ca(OH)_2 + 2\,HNO_3 \longrightarrow Ca(NO_3)_2 + 2\,H_2O$ (1); Double displacement (neutralisation) (1).	1+1+1
16	Antioxidants stop oxidation (1); Nitrogen flushing removes oxygen (1); Refrigeration/airtight containers slow the reaction (1).	1+1+1
17	Green crystals lose water and become whitish (1); Then residue turns reddish brown (1); Smell of burning sulphur (1).	1+1+1
	OR: Right has 10 O atoms (1); $P_4 + 5\,O_2 \longrightarrow P_4O_{10}$ (1); Combination reaction (1).	1+1+1
18	Iron is more reactive than copper (1); Iron displaces copper from $CuSO_4$ (1); Copper cannot displace iron from $FeSO_4$ (1).	1+1+1
	OR: White solid turns grey (1); $2\,AgCl \longrightarrow 2\,Ag + Cl_2$ (1); Photochemical decomposition (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Fe + CuSO4: brown deposit, pale green solution (1); $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ (1); Zn + acid: bubbles of hydrogen (1); $Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$ (1); Reactive metals displace less reactive ones/hydrogen (1).	1+1+1+1+1
	OR: Combination: A + B -> AB, e.g. $C + O_2 \longrightarrow CO_2$ (1); Decomposition: AB -> A + B, e.g. $CaCO_3 \longrightarrow CaO + CO_2$ (1); Displacement: one element replaces another (1); $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$ (1); Double displacement exchanges ions, e.g. $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2\,NaCl$ (1).	1+1+1+1+1
20	Tube with dry air: no rust (1); Boiled water with oil: no rust (no air) (1); Tap water with air: rusts (1); Both air and moisture are needed (1); Prevent by painting, galvanising, oiling (1).	1+1+1+1+1
	OR: Green to white (water lost), then reddish brown (1); $2\,FeSO_4 \longrightarrow Fe_2O_3 + SO_2 + SO_3$ (1); Gases: sulphur dioxide and sulphur trioxide (1); Pungent smell of burning sulphur (1); Thermal decomposition (1).	1+1+1+1+1

Mistakes to avoid:

- Changing the subscripts (formulae) instead of the coefficients while balancing; H_2O must not be written as H_2O_2 to balance oxygen.

- Writing $\mathrm{Mg + O_2 \longrightarrow MgO}$ without balancing, or $\mathrm{MgO_2}$ for magnesium oxide.
- Calling every reaction in which heat is given out 'decomposition'; identify the type from the reactants and products (one product = combination).
- Forgetting that in a displacement reaction the displacing metal must be **more reactive**; $\mathrm{Cu + FeSO_4}$ gives no reaction.
- Mixing up oxidation and reduction: gain of oxygen is oxidation, loss of oxygen is reduction.
- Writing that rusting needs only oxygen; it needs **both** oxygen and moisture. Slaking of lime is exothermic, not endothermic.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Chemical Reactions and Equations** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** loss of water of crystallisation and green colour turning white	Crystals lose 7 H_2O first.
2	**(b)** 2	$Na_2CO_3 + 2\,HCl \longrightarrow 2\,NaCl + H_2O + CO_2$.
3	**(a)** double displacement (precipitation) reaction	Ions exchange and a yellow solid forms.
4	**(b)** exothermic	Heat is released.
5	**(d)** 10	4 x 2 + 2 x 1 = 10.
6	**(d)** $NaOH + HCl \longrightarrow NaCl + H_2O$	Neutralisation involves no gain or loss of oxygen/hydrogen.
7	**(c)** oxidation of fats and oils	Oxidation produces smelly compounds.
8	**(d)** 6	$2\,NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + 2\,H_2O$: 2+1+1+2=6.
9	**(c)** displacement and redox reaction	Zn is oxidised and Cu2+ is reduced.
10	**(d)** It burns with a pop sound	Characteristic test for hydrogen.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Corrosion affects metals; rancidity affects fats/oils in food (1); Corrosion needs moisture and air; rancidity needs oxygen (air) acting on oils (1).	1+1
12	Balance Fe and O: 3Fe and 4H2O (1); $3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$ (1).	1+1
13	$Na_2CO_3 + 2\,HCl \longrightarrow 2\,NaCl + H_2O + CO_2$ (1); Carbon dioxide (1).	1+1
14	$CaCO_3 \longrightarrow CaO + CO_2$ (1); Thermal decomposition (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Balance Cl and H (1); $2\,Al + 6\,HCl \longrightarrow 2\,AlCl_3 + 3\,H_2$ (1); Hydrogen gas (1).	1+1+1
16	Exothermic releases heat, endothermic absorbs heat (1); Example: CH_4 burning (exothermic) (1); Example: decomposition of $CaCO_3$ (endothermic) (1).	1+1+1
17	Beaker becomes hot with hissing (1); $CaO + H_2O \longrightarrow Ca(OH)_2$ (1); Exothermic combination (1).	1+1+1
	OR: Balance Na (1); $H_3PO_4 + 3\,NaOH \longrightarrow Na_3PO_4 + 3\,H_2O$ (1); Double displacement (neutralisation) (1).	1+1+1
18	Painting forms a barrier between iron and air/moisture (1); Galvanisation coats zinc on iron (1); Zinc protects iron even if the layer is scratched (more reactive) (1).	1+1+1
	OR: Hold with tongs (1); Do not look at the flame directly (dazzling light) (1); Clean the ribbon first to remove the oxide layer (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$ (2); (ii) $C_3H_8 + 5\,O_2 \longrightarrow 3\,CO_2 + 4\,H_2O$ (2); (iii) $2\,KClO_3 \longrightarrow 2\,KCl + 3\,O_2$ (1).	2+2+1
	OR: Place both solutions in a flask, with one in a small tube (1); Weigh before mixing (1); Mix: yellow precipitate forms (1); Weigh again: mass is unchanged (1); Law of conservation of mass is verified (1).	1+1+1+1+1
20	$Mg + 2\,HCl \longrightarrow MgCl_2 + H_2$ (1); $2\,KI + Pb(NO_3)_2 \longrightarrow PbI_2 + 2\,KNO_3$ (1); $Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$ (1); $3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$ (1); $CaCO_3 \longrightarrow CaO + CO_2$ (1).	1+1+1+1+1
	OR: Painting/oiling keep out air and moisture (1); Galvanisation: coat of zinc on iron (1); Zinc is more reactive and sacrifices itself (1); Alloying (stainless steel) changes properties (1); Iron mixed with chromium and nickel does not rust (1).	1+1+1+1+1

Mistakes to avoid:

- Changing the subscripts (formulae) instead of the coefficients while balancing; H_2O must not be written as H_2O_2 to balance oxygen.
- Writing $Mg + O_2 \longrightarrow MgO$ without balancing, or MgO_2 for magnesium oxide.

- Calling every reaction in which heat is given out 'decomposition'; identify the type from the reactants and products (one product = combination).
- Forgetting that in a displacement reaction the displacing metal must be **more reactive**; $\mathrm{Cu} + \mathrm{FeSO_4}$ gives no reaction.
- Mixing up oxidation and reduction: gain of oxygen is oxidation, loss of oxygen is reduction.
- Writing that rusting needs only oxygen; it needs **both** oxygen and moisture. Slaking of lime is exothermic, not endothermic.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND